

Topological Analysis and Epidemic Modeling in an Email-Based Social Network

Author: Miriam Martínez Martínez

Master en Física dels Sistemes Complexos i Biofísica.

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We investigate the topology and epidemic dynamics of an email-based social network. Topologically, the network displays an exponential degree distribution, slight positive assortativity, and a high global clustering coefficient with enhanced clustering among low-degree nodes. Dynamically, stochastic and deterministic SIS models agree on the epidemic threshold, which is best predicted by the quenched mean-field estimate.

I. INTRODUCTION

Many real-world systems can be modeled as networks, with nodes representing entities and edges encoding their interactions. Social networks, in particular, provide a natural setting to explore how patterns of human connectivity emerge from individual interactions. Considerable effort has been devoted to characterizing their topology in order to understand how such interactions are organized and constrained [1], [2]. Beyond structural insights, these networks also provide the substrate for studying diffusion phenomena such as the spread of ideas [3] or topics, or even (computer) viruses [4], [5].

One of the simplest models used to study spreading processes on networks is the susceptible–infected–susceptible (SIS) model. In this framework, each node represents an individual that can be in one of two states: susceptible ($n_i(t) = 0$) or infected ($n_i(t) = 1$). Infected nodes recover spontaneously at a rate δ , returning to the susceptible state, while susceptible nodes become infected at a rate λ upon contact with infected neighbors. The dynamics are memoryless and governed by a continuous-time Markov process. To gain analytical insight, a common approach is to employ a mean-field approximation, which assumes independence between the states of neighboring nodes. This leads to a set of coupled ODEs describing the expected infection probability $\rho_i(t)$ of each node i given by

$$\frac{d\rho_i(t)}{dt} = -\delta\rho_i(t) + \lambda(1 - \rho_i(t)) \sum_j a_{ij}\rho_j. \quad (1)$$

where a_{ij} denotes the elements of the adjacency matrix of the network and the sum over j accounts for the influence of all infected neighbors of node i .

In this work, we first examine the structural properties of a real social network and compare them with those from a Configuration Model null model, focusing on degree distribution, clustering, and degree correlations. We then study epidemic spreading on the same network through stochastic SIS simulations, estimating the epidemic threshold and constructing a phase diagram, which we compare with mean-field predictions. Most of the code developed was implemented in *Python*, with the exception of the stochastic SIS dynamics, which was written in *Fortran* to optimize computational efficiency. Additionally, artificial intelligence tools—specifically large

language models—were employed to assist and streamline the coding process.

II. NETWORK DESCRIPTION

The network analyzed in this study is an email communication network from the University Rovira i Virgili (URV) in Tarragona, Spain [6], comprising 1669 users, including faculty, researchers, technical staff, administrators, and graduate students. It is represented as an undirected and unweighted graph, where nodes correspond to email addresses and links indicate mutual communication. In its construction, bulk emails, that is, those sent to more than 50 recipients, were excluded, as well as unidirectional exchanges: two nodes are connected only if both have sent emails to each other. The number of emails exchanged is not considered, meaning that all links are treated equally, regardless of their frequency.

Out of the 1669 nodes in the full network, 1133, connected by 5451 links, belong to the largest connected component, accounting for approximately 68% of all nodes. The remaining nodes are either isolated or form disconnected pairs. In the following analysis, we focus exclusively on this giant component, as it captures the core structure of the network. A visualization of this component is shown in Figure 1.



FIG. 1. **Giant component of the email network.** Visualization generated using a force-directed (spring) layout.

III. TOPOLOGICAL ANALYSIS OF THE NETWORK

To understand how connectivity is organized in the network, we focus on a set of fundamental topological descriptors, including the degree distribution and its cumulative counterpart, along with the average nearest neighbors degree $\langle k_{nn}(k) \rangle$ and the average clustering coefficient $\langle c(k) \rangle$, each evaluated by degree class. To assess the structural relevance of the observed patterns, we compare them with those expected under a suitable null model: the Configuration Model (CM), which generates randomized networks that preserve the degree sequence of the empirical graph while otherwise randomizing the connections. CM is implemented in our analysis considering two different approaches: the standard version, which allows self-loops and multiple edges, and a more constrained variant restricted to simple graphs (CMS), in which such connections are explicitly excluded. This restriction is more appropriate for our context, as self-emails and repeated mutual connections are not represented in the empirical network. Networks under this variant are generated by randomly pairing stubs while rejecting combinations that would result in invalid connections, and repeating the process if necessary until a valid simple graph consistent with the target degree sequence is obtained. For each null model, results are averaged over 100 independent realizations.

The degree distribution $P(k)$ of the email network exhibits an exponential decay (with $\langle k \rangle = 9.62$), as shown in Figure 2 and further supported by the exponential form of its cumulative counterpart $P_c(k)$. Such behavior differs from the heavy-tailed degree distributions observed in other technology-based social networks [7], where power-law scaling gives rise to hubs with extremely high connectivity. The exponential decay found here implies no such high-degree nodes. However, while this is not a truncated power-law in the strict sense, it aligns with the broader idea that real-world social networks deviate from ideal scale-free behavior due to constraints—such as cognitive, temporal, or organizational limits—that restrict the number of active connections an individual can maintain [2]. Since CM is constructed to exactly preserve the degree sequence of the original network, its distributions is not shown.

Another useful metric to explore degree correlations is the average degree of nearest neighbors as a function of node degree, $\langle k_{nn}(k) \rangle$, which reflects whether nodes tend to connect to others with similar degree (assortative mixing) or not (disassortative mixing). The main panel of Figure 3 shows its normalized version, $\langle k_{nn}(k) \rangle / \kappa$, for the empirical network (EN), along with the results obtained from the null models, CM and its simple-graph variant (CMS). Both null models yield nearly flat curves centered around one, as expected from random rewiring that preserves the degree sequence but destroys higher-order structure. The shaded area corresponds to the sample standard deviation across multiple CM realiza-

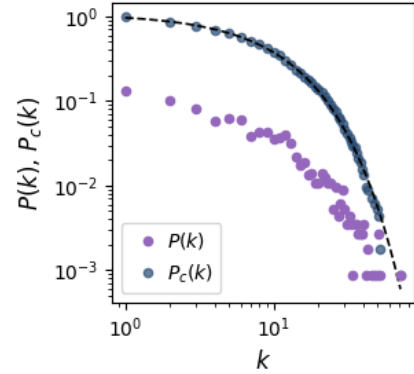


FIG. 2. **Degree distribution $P(k)$ and its cumulative form $P_c(k)$.** The latter follows an exponential decay $P_c(k) \propto e^{-k/k^*}$, with a characteristic degree $k^* = 9.46$ (dashed line). If it would have a power law tail then it would fall on a straight line in a log-log plot.

tions, and serves as a baseline for the level of variation typically observed under such random configurations. In the empirical network, values lie predominantly below one for low-degree nodes and tend to rise above one for higher degrees, forming a broadly increasing trend. While many values remain within the expected range of fluctuations, the shape of the curve suggests a weak assortative tendency. To quantify this tendency more precisely, we compute the degree assortativity coefficient [1] $r = 0.078 \pm 0.017$, with the error estimated using the jackknife method based on edge removal [8]. This value is consistent with the typical assortativity observed in real social networks [9].

The inset panels reveal that, while the CM remains completely flat, CMS exhibits a slight but consistent downward slope—likely a side effect of the structural restrictions imposed during network construction, which reduce the pairing options for high-degree nodes and introduce a mild disassortative bias.

Finally, we compute the average clustering coefficient per degree class, which measures the mean local clustering among nodes of degree k , i.e., the average fraction of connected pairs among a node's neighbors, restricted to nodes with degree k . As shown in the main panel of Figure 4, we again observe a clear discrepancy between EN and both CM and CMS. For the CM-based models, clustering values remain consistently low and nearly flat across all degrees, as expected: the random wiring destroys the local triangle structures on which clustering relies. In contrast, the empirical network exhibits a pronounced decreasing trend: nodes with low degree tend to participate more frequently in triangles, while high-degree nodes show much lower clustering. To further quantify this, we compute the global clustering coefficient for each case: $C_{EN} = 0.2540$, $C_{CM} = 0.0295 \pm 0.0025$, and $C_{CMS} = 0.0287 \pm 0.0021$, where the values for CM and CMS are averages over multiple realizations. Notably,

C_{EN} is almost an order of magnitude larger than the null models. As an additional insight, the CM model shows a slight upward trend compared to CMS in clustering with degree, likely due to the increased probability of multiple edges between high-degree nodes, which artificially raises triangle counts in these configurations.

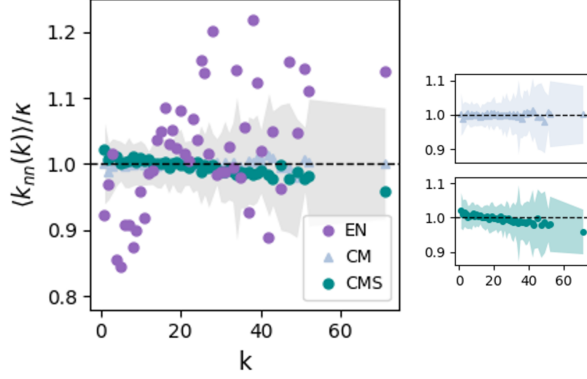


FIG. 3. **Degree correlations in the email network.** Normalized average nearest-neighbor degree $\langle k_{nn}(k) \rangle / \kappa$ for the empirical network (EN), compared to CM and CMS. The shaded area indicates the sample standard deviation over CM realizations. Insets show CM and CMS behaviors in greater detail with shaded bands representing their respective standard deviations across realizations.

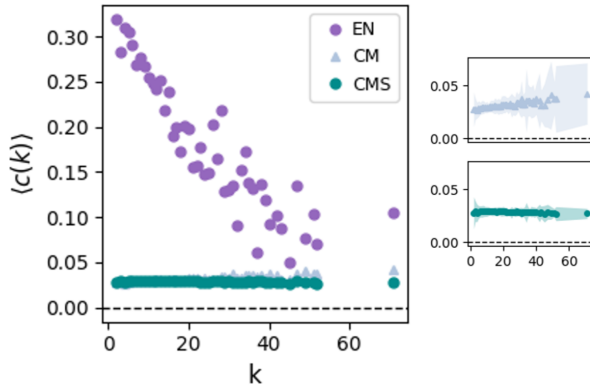


FIG. 4. **Clustering as per degree class.** Average clustering coefficient $\langle c(k) \rangle$ per degree class for the empirical network (EN), CM, and CMS. The insets show CM and CMS in isolation; shaded areas represent the sample standard deviation across realizations.

IV. SPREADING DYNAMICS UNDER THE SIS MODEL

We now turn to the study of SIS dynamics on the network to better understand how an infectious process spreads through its structure. The network can be interpreted as a community of N individuals connected by email interactions, where each node represents a person and each link reflects a potential exposure to others. To

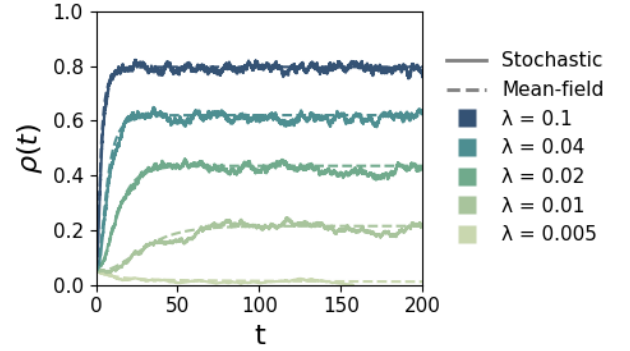


FIG. 5. **Time evolution of the SIS model.** Prevalence $\rho(t)$ obtained from stochastic simulations (solid lines) and mean-field approximation (dashed lines) for different values of the infection rate λ .

capture the temporal unfolding of e-mail virus propagation, we adopt two complementary approaches: (i) a stochastic simulation based on discrete infection and recovery events, which tracks the time evolution of the infected population; and (ii) a deterministic description in terms of coupled differential equations that estimate the infection probability of each node under a heterogeneous mean-field approximation.

To simulate the stochastic evolution of this state over time, we employ the Gillespie algorithm. In the context of the SIS model, two types of events are considered: recovery events, which occur with probability proportional to $\delta N_I(t)$, where $N_I(t)$ is the number of infected nodes at time t ; and infection events, which occur with probability proportional to $\lambda E_{\text{act}}(t)$, where $E_{\text{act}}(t)$ is the number of active edges, i.e., links between infected and susceptible nodes. Based on these probabilities, the algorithm determines both the type and timing of the next event. From a computational perspective, this is implemented using dynamic lists that store the current set of infected nodes and active edges, which are updated as the simulation progresses. On the deterministic side, the system of coupled differential equations are numerically solved using the `solve_ivp` function from SciPy, employing the Runge-Kutta method of order 5(4), i.e., RK45.

The simulations have been carried out with a recovery rate fixed at $\delta = 0.1$, which sets the timescale of the process by controlling how quickly infected individuals return to the susceptible state, and an initial condition of 57 infected nodes, corresponding to approximately 5% of the network. Figure 5 shows the time evolution of the prevalence $\rho(t)$ for different values of the infection rate λ . In all cases, the infection spreads and stabilizes around a steady-state value, as predicted by the mean-field approximation (dashed lines) and the stochastic simulations (solid lines) fluctuate around these values. Remarkably, for the lowest infection rate considered ($\lambda = 0.005$), the epidemic dies out entirely in the stochastic simulation, reaching the absorbing state with no infected nodes.

In the context of epidemic spreading, a central ques-

tion is to determine under which conditions the infection can sustain itself and propagate through the network. This is typically characterized by the epidemic threshold, defined as the critical value of the ratio $(\lambda/\delta)_c$ between the infection and recovery rates. This threshold separates two distinct regimes: for $\lambda/\delta < (\lambda/\delta)_c$, the infection eventually dies out, while for $\lambda/\delta > (\lambda/\delta)_c$, it can persist in the population, leading to an endemic state. To explore this transition, we construct the phase diagram of the SIS model by evaluating the steady-state prevalence for several values of λ/δ , keeping the recovery rate fixed at $\delta = 0.1$ and starting each simulation with the same number of initially infected nodes. For each parameter setting, 100 independent realizations of the stochastic spreading process were performed, and the average fraction of infected nodes at time $t = 200$ —sufficiently long for the system to reach a stationary state—was computed. The steady state values of the mean-field approximation were also computed.

The resulting diagram, shown in Figure 6, reveals in fact how the stationary prevalence ρ_{st} transitions from zero to positive values, indicating this regime shift. The inset highlights the critical region where the transition occurs, allowing the epidemic threshold to be visually identified. Additionally, the figure shows theoretical predictions for the threshold computed under different analytical approaches: the quenched mean-field approximation, which explicitly incorporates the full network structure, yields $(\lambda/\delta)_c = 1/\Lambda_{\max}^A$, where $\Lambda_{\max}^A$ is the largest eigenvalue of the adjacency matrix; the heterogeneous mean-field approximation, based on the degree distribution and assuming statistical independence between nodes, gives $(\lambda/\delta)_c = \langle k \rangle / \langle k^2 \rangle$; and finally, a simplified estimate based on homogeneous mixing assumes all nodes are statistically equivalent and results in $(\lambda/\delta)_c = 1/\langle k \rangle$. Among these, the former yields $(\lambda/\delta)_c = 0.0482$, which is the closest to the transition point observed in the stochastic simulations; the second gives $(\lambda/\delta)_c = 0.0535$, slightly overestimating it; and the latter, with $(\lambda/\delta)_c = 0.1039$, significantly overestimates the actual threshold. This better agreement is expected, as the quenched mean-field approach incorporates the full network topology through the spectral properties of the adjacency matrix.

V. CONCLUSIONS

In this work, we analyzed the structure of an email-based social network from the University Rovira i Virgili, focusing on key statistical measures including degree distribution, assortativity, and clustering coefficient. Our topological analysis revealed three main findings: (1) the degree distribution decays exponentially, (2) the network exhibits a weak but positive assortativity ($r \approx 0.078$), and (3) a high global clustering coefficient ($C \approx 0.254$) with a clear inverse relationship between node degree and local clustering (i.e., lower-degree nodes participate in triangles more frequently). Taken together, these results

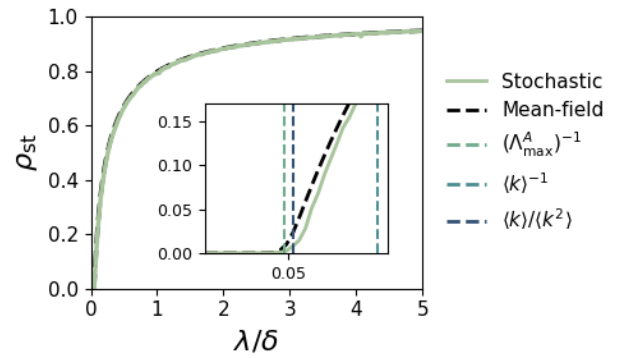


FIG. 6. **Phase diagram of the SIS model.** Stationary prevalence ρ_{st} as a function of the control parameter λ/δ , obtained from stochastic simulations (solid line) and mean-field approximation (dashed line). The inset highlights the critical region, with vertical lines indicating theoretical thresholds predicted by different analytical approaches.

imply that the network lacks dominant hubs, tends to connect nodes of similar degree modestly more than random expectation, and forms tightly knit local communities.

Turning to epidemic dynamics, we employed both stochastic Gillespie simulations and a deterministic mean-field approximation of the SIS model, finding close agreement between the two approaches across a range of infection rates. The phase diagram shows a clear transition from disease extinction to endemic states. Notably, the epidemic threshold estimated via the quenched mean-field formula that yields $(\lambda/\delta)_c = 0.0482$, aligns most closely with our simulation results, outperforming the heterogeneous mean-field estimate, and the homogeneous-mixing approximation.

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